

# REPRODUCIBLE MATHEMATICA AND LEAN VERIFICATION FOR TRACE-EUCLIDEAN-V9

## COMPUTATIONAL COMPANION AND MAINTENANCE GUIDE

ABSTRACT. This document records the reproducible Mathematica checks and selected Lean 4 proof obligations associated with `Trace-Euclidean-v9.tex`. It records the SHA-256 digest of the private manuscript version used to prepare the extracted inputs, maps each mathematical component to a named code module and stable internal LaTeX anchor, and imports all numerical results from machine-readable output. In the pages below, those anchors are rendered as the theorem, equation, section, and table numbers in the compiled manuscript, together with page numbers where they help direct comparison. Exact identities, certified interval calculations, finite enumerations, and finite diagnostics are kept distinct. The accompanying workflow is designed so that a later manuscript revision requires changes only in the affected manifest entry or check module; the result tables in this document are then regenerated automatically. The public repository intentionally excludes both the manuscript TeX source and the rendered manuscript PDF.

### 1. PURPOSE AND EVIDENCE BOUNDARY

The package checks the formulas and finite computations used in the manuscript. A record marked **PASS** establishes only the scope stated for that check. In particular, a finite diagnostic does not prove a theorem for all number fields or all lattice vectors, and a high-precision root approximation does not certify existence or uniqueness. General compactness, monotonicity, finiteness, and source-dependent assertions remain mathematical arguments in the manuscript.

The evidence classes used here are:

- (i) an *exact symbolic identity*, simplified with exact input;
- (ii) a *certified rational interval*, whose endpoints are exact rational numbers and whose sign excludes zero;
- (iii) a *finite enumeration*, over the complete rectangle recorded below;
- (iv) a *high-precision approximation*, used only to check printed decimals;
- (v) a *finite diagnostic*, explicitly limited to the configured sample.

Publication boundary. The repository-level publication check permits only explicitly listed verification-document `.tex` and `.pdf` files. It rejects the private source-snapshot and manuscript-build directories, complete TeX or PDF documents outside that allowlist, and opaque archives. Consequently, the recorded source digest identifies the manuscript version used for extraction without distributing the manuscript source or its rendered PDF.

### 2. VERSION BINDING AND RECORDED RUN

The private maintainer extraction step computes the SHA-256 digest of the manuscript snapshot. During that build, the Mathematica driver recomputes the digest before loading any mathematical module, so a changed snapshot cannot silently reuse stale inputs. The public repository does not contain that snapshot. A public run therefore reads the

recorded digest from `inputs/manuscript_inputs.json`, checks its form and agreement with the result metadata, and then reruns the published calculations. This permits independent reproduction of the computations without distributing the manuscript itself.

| Item                          | Recorded value                                                                |
|-------------------------------|-------------------------------------------------------------------------------|
| Private source identifier     | <code>Trace-Euclidean-v9.tex</code>                                           |
| SHA-256                       | <code>a2f522d077c600e0dc747dcaa8b06ba4e31ecd0b49b83dae468c9b61915a0e99</code> |
| Source lines                  | 2372                                                                          |
| Wolfram kernel                | 15.0.0 for Microsoft Windows (64-bit) (May 15, 2026)                          |
| Completed                     | 2026-09-09T22:40:58                                                           |
| PASS / WARN / FAIL            | 2165 / 0 / 0                                                                  |
| Certified classical pairs     | 24                                                                            |
| Certified integral pairs      | 112                                                                           |
| Square-free parameters tested | 121                                                                           |
| Root precision                | 80 digits                                                                     |
| Rational interval output      | 40 decimal places, rounded outward                                            |
| Kernel time                   | 14.632 seconds                                                                |

The status convention is as follows. A **FAIL** is a failed exact, certified, coverage, or integrity check and gives batch exit code 1. A **WARN** records a manuscript presentation or rounding issue while the verification engine remains internally consistent and gives exit code 2. A run with neither gives exit code 0.

### 3. SOURCE, CODE, AND RESULT CORRESPONDENCE

Every test record in `results/tests.json` and `results/tests.csv` contains the fields `id`, `module`, `anchor`, `evidence`, `category`, `status`, `actual`, `expected`, and `note`. Thus a result can be traced in both directions: from a manuscript label to a code module and its test records, or from a test ID back to the relevant manuscript location. Stable LaTeX labels remain internal machine keys; the manual displays the corresponding compiled references, such as “Lemma 4.6,” rather than raw label names. Table checks additionally use stable row IDs such as `admissible-classical:r01`; source line numbers are retained only as a convenience and are not used as identifiers.

#### Analytic functions, derivatives, and extrema

**Code:** `checks/analytic_bounds.wl`. **Result (P/W/F):** 134 / 0 / 0.

**Manuscript locations:** Equation (4.1), Lemma 4.1, Lemma 4.2, Lemma 4.3, Lemma 4.4, Lemma 4.5, Lemma 4.6, Lemma 4.7, Lemma 4.8, Lemma 4.9.

**Evidence:** exact symbolic identities, certified rational intervals, and high-precision approximations.

#### Admissible-pair tables and excluded tails

**Code:** `checks/admissible_tables.wl`. **Result (P/W/F):** 1677 / 0 / 0.

**Manuscript locations:** Table 1, Table 2, Table 3, Table 4.

**Evidence:** certified finite enumeration plus analytic tail conditions from the manuscript.

#### Trace Gram matrices and quadratic-field Voronoi cells

**Code:** `checks/trace_geometry.wl`. **Result (P/W/F):** 272 / 0 / 0.

**Manuscript locations:** Lemma 3.1, Proposition 6.1, Equation (6.2), Theorem 1.8.

**Evidence:** exact algebra, exact Voronoi vertices, and bounded finite diagnostics.

**Gamma-function boundary diagnostics**

**Code:** checks/gamma\_diagnostics.wl. **Result (P/W/F):** 81 / 0 / 0.

**Manuscript locations:** Equation (3.12), Equation (3.13).

**Evidence:** finite high-precision diagnostics; the cited analytic inequalities remain external inputs.

The machine-readable version of this table is `results/verification_ledger.json`.  
The detailed result count by check category is:

| Check category              | Total | PASS | WARN | FAIL |
|-----------------------------|-------|------|------|------|
| Gamma finite sample         | 81    | 81   | 0    | 0    |
| certified table upper bound | 42    | 42   | 0    | 0    |
| exact Voronoi               | 13    | 13   | 0    | 0    |
| exact algebra               | 58    | 58   | 0    | 0    |
| exact classification        | 1     | 1    | 0    | 0    |
| exact implication           | 2     | 2    | 0    | 0    |
| exact inequality            | 5     | 5    | 0    | 0    |
| exact integer proof         | 2     | 2    | 0    | 0    |
| exact trace Gram            | 8     | 8    | 0    | 0    |
| finite diagnostic           | 121   | 121  | 0    | 0    |
| finite enumeration          | 1525  | 1525 | 0    | 0    |
| finite field verification   | 121   | 121  | 0    | 0    |
| integrity                   | 1     | 1    | 0    | 0    |
| manuscript approximation    | 15    | 15   | 0    | 0    |
| rational certificate        | 46    | 46   | 0    | 0    |
| source binding              | 9     | 9    | 0    | 0    |
| symbolic limit              | 3     | 3    | 0    | 0    |
| symbolic sign               | 10    | 10   | 0    | 0    |
| table approximation         | 34    | 34   | 0    | 0    |
| table coverage              | 2     | 2    | 0    | 0    |
| table display precision     | 34    | 34   | 0    | 0    |
| table maxima                | 24    | 24   | 0    | 0    |
| table outward interval      | 8     | 8    | 0    | 0    |

## 4. ANALYTIC DEFINITIONS AND EXTREMA

The code uses the manuscript definitions

$$\begin{aligned}
 a &= \frac{1}{6}, & b &= \frac{e^2}{2\pi} - 1, \\
 K(x) &= \log(2\pi) - \log x - 1, & g(x, y) &= x \log(2\pi(y + b)) - \log(\pi(xy + 2a)) + xyK(x), \\
 g_s(x, y) &= \frac{g(x, y)}{x}, & g_n(x, y) &= y \log 2 + \frac{g(x, y)}{x}.
 \end{aligned}$$

The module first compares exact symbolic derivatives with the displayed formulas. It then checks the signs needed at the stated thresholds by rational interval arithmetic. The roots returned by `FindRoot` at 80-digit working precision are used only for the following comparison with decimals printed in the manuscript.

| Claim ID       | Manuscript location | Line | Printed | Computed      | Status |
|----------------|---------------------|------|---------|---------------|--------|
| b              | Section 4           | 1098 | 0.176   | 0.1760048029  | PASS   |
| y0             | Lemma 4.1           | 1128 | 25.194  | 25.1938160912 | PASS   |
| y1             | Lemma 4.1           | 1164 | 5.7     | 5.6579939435  | PASS   |
| x0             | Lemma 4.2           | 1174 | 4.375   | 4.3745521287  | PASS   |
| z-minus        | Lemma 4.2           | 1200 | 0.477   | 0.4765865552  | PASS   |
| z-plus         | Lemma 4.2           | 1200 | 0.699   | 0.6994182477  | PASS   |
| g-maximum      | Lemma 4.3           | 1247 | 3.329   | 3.3286422771  | PASS   |
| x-boundary     | Lemma 4.3           | 1249 | 5.253   | 5.2534385501  | PASS   |
| y-rank3        | Lemma 4.6           | 1365 | 2.347   | 2.3471186383  | PASS   |
| gs-max         | Lemma 4.6           | 1363 | 1.104   | 1.103803611   | PASS   |
| two-pi-over-e  | Lemma 4.6           | 1375 | 2.311   | 2.3114546996  | PASS   |
| xn             | Lemma 4.8           | 1391 | 9.803   | 9.8031890569  | PASS   |
| gn-max         | Lemma 4.9           | 1460 | 2.361   | 2.3610934112  | PASS   |
| yn-rank5       | Lemma 4.9           | 1462 | 9.999   | 9.9985918124  | PASS   |
| four-pi-over-e | Lemma 4.9           | 1470 | 4.623   | 4.6229093992  | PASS   |

For the global comparison in Lemma 4.3, the computation certifies

$$2.346 < y_0(3) < 2.348, \quad 5.252 < x_0(1) < 5.254, \\ g(3, y_0(3)) < 3.325 < 3.328 < g(5.253, 1).$$

The Hessian identities are checked symbolically. The argument that the maximum is attained and that no point at infinity is omitted remains the analytic argument in the manuscript.

## 5. COMPLETE INVENTORY OF SECTION 4 CALCULATIONS

The following ledger lists every calculation in Section 4 that can be checked directly by the present Mathematica methods: definitions and symbolic derivatives, algebraic reductions, interval-certified signs and root brackets, threshold evaluations, Hessian tests, boundary comparisons, and the stated one-variable limits. Each row names the exact location in the compiled manuscript and the evidence used by its associated test records.

| Manuscript location   | Calculation checked                                                                                                                         | Evidence                                                                    | Result    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------|
| Section 4, p. 13      | The printed value of $b=e^2/(2\pi)-1$ .                                                                                                     | 80-digit evaluation and comparison with the printed decimal                 | PASS (1)  |
| Equation (4.1), p. 14 | The formulas for $g$ , $g_s$ , and $g_n$ , all five displayed first and second partial derivatives, and the substitution leading to $G_t$ . | exact symbolic identities plus rational certificates for $0 < b < 1/4 < 2a$ | PASS (12) |
| Lemma 4.1, p. 14      | The factorization, intermediate square identity, positive coefficient, and strict negativity of the second $x$ -derivative of $g$ .         | exact symbolic identities and quantified sign check                         | PASS (4)  |
| Lemma 4.1, p. 14      | The definition of $f(y)=\partial_x g(1,y)$ , its first two derivatives, and the expression of $f''$ through $h$ .                           | exact symbolic identities                                                   | PASS (4)  |
| Lemma 4.1, p. 14      | The derivatives of $h$ , the three displayed constant identities, and positivity of $h'(1)$ and $h(1)$ .                                    | exact identities, quantified sign check, and outward rational intervals     | PASS (9)  |

| Manuscript location | Calculation checked                                                                                                                                         | Evidence                                                                                     | Result    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------|
| Lemma 4.1, p. 14    | The displayed values $f'(1)$ and $f(1)$ , their signs, the limiting sign of $f'$ , and the limits of $f$ and $\text{partial}_x g$ .                         | exact identities, symbolic limits, and outward rational intervals                            | PASS (7)  |
| Lemma 4.1, p. 14    | The roots $y_1$ and $y_0$ , including strict rational brackets and the decimals printed in the manuscript.                                                  | outward rational root brackets and 80-digit approximations                                   | PASS (6)  |
| Lemma 4.2, p. 15    | The formula $h=\text{partial}_y g/x$ , its $y$ -derivative, its strict negative sign, and its limit $K(x)$ .                                                | exact identities, quantified sign check, and symbolic limit                                  | PASS (4)  |
| Lemma 4.2, p. 15    | The critical-point formula after $x=z^2$ , its equivalent quadratic inequality, and the two printed quadratic roots below 1.                                | exact identities, rational endpoint sign, and 80-digit approximations                        | PASS (7)  |
| Lemma 4.2, p. 15    | The threshold $K(2\pi/e)=0$ , the inequality $2\pi/e>1$ , monotonicity of $K$ , and positivity of the reciprocal difference used in case (i).               | exact identities, quantified sign check, and outward rational interval                       | PASS (5)  |
| Lemma 4.2, p. 15    | The definition and derivative of $f(x)=h(x,1)$ , its strict decrease and limit, the positivity at $2\pi/e$ , and the unique threshold $x_0$ .               | exact identities, symbolic sign and limit, rational root bracket, and 80-digit approximation | PASS (8)  |
| Lemma 4.3, p. 16    | The critical-point identity, strict negativity of $f'$ , both differentiated identities at a critical point, and the deduction $g_{xy}<0$ from $g_{xx}<0$ . | exact symbolic identities, quantified polynomial sign, and a quantified real implication     | PASS (6)  |
| Lemma 4.3, p. 16    | The Hessian determinant reduction and the strict negative sign at every interior critical point.                                                            | exact symbolic reduction and a quantified real implication                                   | PASS (2)  |
| Lemma 4.3, p. 16    | Both boundary equations, their root brackets, and the certified strict comparison selecting $g(x_0(1),1)$ .                                                 | exact identities, outward rational intervals, and printed-value checks                       | PASS (11) |
| Lemma 4.3, p. 16    | The two tail limits, $K(3)<0$ , and the monotonic decrease of $xK(x)$ for $x\geq 3$ used to obtain a uniform $y$ -tail bound.                               | exact symbolic limits and derivatives, quantified sign, and outward rational intervals       | PASS (7)  |
| Lemma 4.4, p. 17    | The $x$ -derivative of $g_s$ , the reduction to $f_s(z)$ , the derivative of $f_s$ , $z\geq 4/3$ , positivity of $3z^2-3z+1$ , and $f_s(4/3)<0$ .           | exact identities, quantified inequalities, and outward rational interval                     | PASS (6)  |
| Lemma 4.5, p. 18    | The identity $\text{partial}_y g_s=(\text{partial}_y g)/x$ transferring the signs and zeros from Lemma 4.2.                                                 | exact symbolic identity                                                                      | PASS (1)  |
| Lemma 4.6, p. 18    | The ordering $2\pi/e<3<x_0$ needed at $x=3$ , together with the printed boundary point and maximum of $g_s$ .                                               | outward rational interval and 80-digit approximations                                        | PASS (6)  |
| Lemma 4.7, p. 18    | The identity $\text{partial}_x g_n=\text{partial}_x g_s$ transferring strict decrease in $x$ .                                                              | exact symbolic identity                                                                      | PASS (1)  |
| Lemma 4.8, p. 18    | The first and second $y$ -derivatives of $g_n$ and the limiting value $\log(4\pi/(xe))$ .                                                                   | exact symbolic identities and symbolic limit                                                 | PASS (3)  |
| Lemma 4.8, p. 18    | The definition and derivative of $f_n(x)$ , strict decrease and limit, positivity at $4\pi/e$ , and the unique threshold $x_n$ .                            | exact identities, symbolic sign and limit, rational root bracket, and 80-digit approximation | PASS (9)  |

| Manuscript location | Calculation checked                                                                      | Evidence                                               | Result   |
|---------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------|----------|
| Lemma 4.9, p. 20    | The ordering $4\pi/e < 5 < x_n$ , the root $y_n(5)$ , and the printed maximum of $g_n$ . | outward rational intervals and 80-digit approximations | PASS (7) |

This is a computational coverage statement, not a replacement for the proof. Arguments that pass from these calculations to global existence, uniqueness, or exhaustion of all cases still depend on the hypotheses and deductions in the manuscript. The ledger makes that boundary visible while ensuring that no computational test in the Section 4 analytic module is left unassigned.

## 6. CERTIFIED INTERVAL CONSTRUCTION

The interval implementation uses exact rational endpoints. It bounds  $\pi$  through Machin's identity

$$\pi = 16 \arctan\left(\frac{1}{5}\right) - 4 \arctan\left(\frac{1}{239}\right)$$

and an alternating arctangent series through exponent 121. It bounds  $e$  by the sum through  $1/100!$  together with an explicit geometric remainder. For a positive rational number, powers of 2 first reduce the logarithm argument to  $[1, 2]$ ; with  $u = (r-1)/(r+1)$ , the code uses

$$\log r = 2 \sum_{j=0}^{47} \frac{u^{2j+1}}{2j+1} + R, \quad 0 \leq R \leq \frac{2u^{97}}{97(1-u^2)}.$$

Every arithmetic operation is rounded outward to 40 decimal places, while the stored endpoints remain rational numbers. Consequently, a certified positive or negative sign does not depend on floating-point evaluation.

## 7. ADMISSIBLE-PAIR TABLES

For the classical case the complete rectangle is  $3 \leq n \leq 12$ ,  $1 \leq d \leq 10$ ; for the integral case it is  $5 \leq n \leq 29$ ,  $1 \leq d \leq 57$ . The computation classifies every integer point by a rational interval for  $g_s(n, d)$  or  $g_n(n, d)$ , and compares the positive set with the rows extracted from the two admissible-pair tables. The tail signs at  $(13, 1)$ ,  $(3, 11)$ ,  $(30, 1)$ , and  $(5, 58)$  are certified separately. Exhaustion outside the two rectangles also uses the monotonicity lemmas in the manuscript.

For a table range, its lower endpoint is required to be rounded downward and its upper endpoint upward. A number described as an upper bound is checked by certifying  $g(n, d) < \log B$ . The current run produced the following records that are not **PASS**:

No **WARN** or **FAIL** record was produced by this run.

If a future row is reported here, its stable row ID identifies the extracted table entry. For a rounding warning, the final column gives the smallest decimal at the same precision that the interval computation certifies as an outward upper bound. The present values in Table 1 have been adjusted by this rule.

## 8. TRACE GRAM AND VORONOI CALCULATIONS

The trace-Gram module tests non-diagonal binary Gram matrices together with distinct fractional principal ideals over several real quadratic fields. All entries and determinants are reduced exactly modulo  $z^2 - m$ . These tests check the determinant formula on deliberately nontrivial inputs; the general formula remains the symbolic proof in Lemma 3.1.

For the hexagonal case, the code solves the adjacent bisector equations exactly and verifies all six displayed facet inequalities. If

$$G = \begin{pmatrix} p & r \\ r & q \end{pmatrix},$$

then the computed vertex norm is simplified to

$$\frac{pq(p + q - 2r)}{4(pq - r^2)}.$$

After substituting the real-quadratic Gram matrix, each vertex has squared norm  $(m + 1)^2/(8m)$ . The additional enumeration over square-free  $2 \leq m \leq 200$  is a finite diagnostic; completeness of the Voronoi facets for arbitrary  $m$  comes from the manuscript's geometric argument.

## 9. LEAN FORMALIZATION AND SEMANTIC BOUNDARY

The directory `lean/` is a Lake project pinned to Lean 4.32.1 and a fixed mathlib revision. Its modules check the strict and non-strict covering-radius implications, the power-mean witness transfer, cancellation of the factor-two scaling, the corrected Hessian-saddle step in Lemma 4.3, finite-parameter assembly, the two-dimensional vertex algebra above, and the quadratic candidate calculations.

The formal source contains no `sorry`, `sorryAx`, project `axiom`, `native_decide`, `run_tac`, `unsafe`, `extern`, or `implemented_by`. The public endpoint audit prints only `propext`, `Classical.choice`, and `Quot.sound`. These are standard Lean and mathlib logical dependencies.

The formalization is intentionally partial. In particular, it does not yet construct the totally real fields and their lattices, formalize the full Voronoi-cell argument, or state the complete finiteness and quadratic-field classification theorems as Lean endpoints. Thus successful compilation certifies the encoded components, not the complete manuscript. The paper-to-code map and the English semantic-fidelity audit are in `THEOREM_INDEX.md` and `docs/audit/`.

## 10. REPRODUCTION

The public code, extracted inputs, results, and documentation are available at <https://github.com/hzldew-git/PaperVerification/tree/main/Trace-Euclidean>. To rerun the published calculations from the package root, use:

```
wolframscript -file run_verification.wls
python tools/build_report.py
python tools/check_delivery.py
```

For the formalized components, change to `lean/` and run:

```
lake exe cache get
lake build
lake env lean TraceEuclideanTest/MainTheoremAudit.lean
```

No manuscript file is required for these commands. To inspect variables interactively, evaluate

```
Get[".../verify_trace_euclidean_v9.m"]
```

in Mathematica. The `.m` file deliberately does not call `Exit`; only `run_verification.wls` turns the recorded status into a batch exit code.

After a manuscript revision, the maintainer keeps the manuscript outside the Git repository and runs the private refresh pipeline with:

```
.\build_all.ps1 -ManuscriptPath 'D:\private\path\Trace-Euclidean-v9.tex'
```

The temporary source snapshot and compiled manuscript are written only to ignored private-work directories. The pipeline regenerates the extracted inputs, displayed manuscript numbering, result files, and this verification manual, then invokes the repository manuscript-exclusion check. The optional executable parameters may be used when Python, WolframScript, or TeX Live is installed elsewhere. No network connection or third-party Wolfram package is required for the calculation run.